

Narayan Industrial Township (NIT): A Systems-Led Recovery and AI-Driven Infrastructure Stabilization Strategy

Fluxion Candidate Assessment – Submission

Role Applied For: Data Science Engineer (Operational Analytics, Predictive Modelling & Reporting)

Discipline/Background: Applied Statistics and Informatics at IIT Bombay, with experience in machine learning, statistical modeling, time-series forecasting and predictive systems.

1 A Specific Experience That Genuinely Interested Me

During my Telco Churn Prediction project, I initially treated it like a normal classification assignment — clean the data, train models, compare accuracy, and move on. But one thing kept bothering me: some models showed high accuracy while still missing many actual churn customers.

I remember checking the confusion matrix for the first time and realizing that predicting “non-churn” for almost everyone could still look good numerically because most customers stayed. That was the moment the project stopped feeling like a textbook problem.

I started focusing more on recall and ROC-AUC instead of just accuracy, because in a real telecom business, missing a customer who is about to leave matters more than being generally correct. Tuning models with that mindset made the work feel practical and real, and that shift is what genuinely got me interested in machine learning.

2 SCENARIO UNDERSTANDING

Core systemic problem: NIT is a coupled infrastructure–manufacturing–human system operating at 34% capacity with monthly losses of 7 crore. Multiple failures coexist, but the critical observation is that no single metric is an outlier in isolation—rather, the *propagation paths* between them have broken down. The township’s design assumed stable power (99.2% uptime) as a foundation, but actual uptime (71%) destabilizes everything downstream.

Operational symptoms (categorised):

- *Production:* 5/12 light manufacturing units operational; 3 shut down; 4 never launched.
- *Logistics:* 1/3 warehouses active; one operator exiting.

- *Workforce*: 67% attrition vs 18% industry norm.
- *Infrastructure*: Power 71% uptime; water supply failure events (0–223); connectivity degraded (3G vs 4G).
- *Medical*: 340% occupancy during shift changes (designed 120 beds).
- *Financial*: 11cr revenue vs 18cr cost; 45cr budget for 18 months.

Structural weaknesses (hypothesised):

- Power is a *keystone dependency* but lacks real-time monitoring for root cause isolation.
- Data backbone exists but is not producing actionable visibility (evidenced by degraded connectivity in Zone D and contractual dispute with telecom provider).
- Workforce attrition is both an outcome and a driver, yet no leading indicators are tracked.
- Medical centre overload suggests a mismatch between scheduled shift patterns and health-care demand timing—a coordination failure, not capacity failure.

Missing information (critical unknowns):

- Distribution of equipment downtime across units (concentrated vs uniform).
- Actual power interruption characteristics (frequency, duration, magnitude, which zones).
- Workforce departure reasons (voluntary vs involuntary, skill levels, tenure).
- Water supply failure pattern (seasonal, infrastructure-specific, or operator error).

3 PROBLEM FRAMING

3.1 Eight Load-Bearing Questions

#	Question	Why the answer would change my approach
1	What is the <i>actual distribution</i> of equipment downtime across the 5 active manufacturing units (mean, variance, tail)?	If downtime is concentrated in 1–2 units, the problem is local (specific equipment or operator). If uniform, it’s systemic (power, maintenance process, or workforce skill).
2	For each power failure event, what is the <i>duration</i> and <i>advance warning time</i> (if any)?	Short, frequent outages (<5 min) cause cumulative equipment wear and production loss but no evacuation. Long outages (>2 hours) drive attrition directly. Warning time determines whether predictive maintenance can mitigate.
3	What is the <i>lag</i> between a power outage and a worker’s decision to leave?	If lag <1 week, attrition is reactive to immediate frustration. If lag >1 month, attrition reflects accumulated instability—different intervention (immediate vs structural).

4	Of the 4 units that never launched, how many had <i>completed infrastructure hookups</i> (power, water, data) at the time of decision?	If hookups were incomplete, the barrier is infrastructure availability. If complete but they still didn't launch, the barrier is confidence in <i>reliability</i> —requires reputation repair, not just physical fixes.
5	What is the <i>actual water consumption pattern</i> during failure events (pre-failure baseline vs post-failure)?	Reveals whether failures are predictable (seasonal, demand-driven) or random. Predictable failures can be managed by scheduling and buffering; random failures require redundancy.
6	What is the <i>census of the medical centre during non-shift-change hours</i> ?	If occupancy drops to <100% outside shift changes, the problem is peak demand management (scheduling). If it remains >200%, there is a chronic population underestimation or lack of alternative care.
7	What is the <i>telecom dispute specifically about</i> (SLA penalty clauses, performance measurement methodology, or payment)?	If dispute is about measurement (e.g., how uptime is defined), then data infrastructure is the root. If about payment, it's a governance issue—different resolution path.
8	Of the 3 units that shut down after attempting operations, what was the <i>sequence of failures</i> in their final month (power → equipment → quality → order cancellation)?	Determines whether shutdown was sudden (catastrophic failure) or gradual (eroding margins). Sudden suggests a single unmanageable shock; gradual suggests systemic unviability.

3.2 Single Best Hypothesis

Hypothesis: Unreliable power (71% uptime) is the primary cause, not because it directly stops production, but because it generates *unpredictable and frequent disruptions* that cascade into equipment damage (34% downtime), logistics unreliability (110 vs 600 trucks/day), workforce attrition (67%), and revenue collapse—while also deterring new units from launching and driving existing operators to exit.

Confidence level: Medium.

Why not higher: The data does not specify power failure *characteristics* (duration, frequency, advance warning, zonal distribution). It is possible that power is a symptom, not a root—for example, water failures could be causing power trips (pumps failing), or workforce skill loss could be causing improper equipment handling that appears as power-related downtime. Without time-stamped event sequences, causality is inferred from design logic, not proven.

3.3 Two Suspicious Patterns

Pattern	Why it is suspicious
“Average equipment downtime per active manufacturing unit: <5% designed vs 34% actual”	The average hides distribution. If one unit has 80% downtime and four have 10% each, the average is 24%—but the average here is 34%. That implies either (a) all five units are uniformly failing (34% each), which is unlikely unless a shared system (power) is failing uniformly, or (b) the “active” denominator is mis-specified (units counted as active despite zero production). The phrase “per active manufacturing unit” is ambiguous: does “active” mean “connected to power” or “producing output”? If the latter, downtime definition is circular.
“Medical center occupancy: 340% during shift changes”	With 120 beds designed, 340% implies 408 patients simultaneously. Total worker housing population is 4,200 residents. A shift-change surge of 408 patients means nearly 10% of the <i>entire resident population</i> requires acute care at the same time twice daily. That is epidemiologically implausible for a healthy working-age population unless: (a) the medical centre is serving non-residents (e.g., workers from outside, or families not counted), (b) “occupancy” is measuring <i>visits per hour</i> rather than <i>bed occupancy</i> (units mismatch), or (c) there is a severe occupational hazard (e.g., chemical exposure, heat stroke) concentrated at shift changes.

4 SYSTEM DECOMPOSITION

4.1 NIT as a System – Dependency Map (Textual Form)

I model NIT as a directed acyclic graph of *resource flows* and *information dependencies*, with feedback loops.

Tier 1 (Foundation – no redundancy):

- **Power substation (Zone D)** → supplies Zones A, B, C, D.
- **Water treatment (Zone D)** → requires power. Fails if power fails.
- **Data centre (Zone D)** → requires power and cooling (water). Provides visibility across all zones.

Tier 2 (Direct users):

- **Manufacturing units (Zone A)** → require power (continuous), water (process cooling), data (inventory/order mgmt), and skilled workers.
- **Logistics hub (Zone B)** → requires power (gates, cranes, cold storage), data (tracking), and manufacturing output to move.

- **Medical centre (Zone C)** → requires power (lights, equipment, refrigeration), water, data (records), and workers (as patients).

Tier 3 (Human capital loop – reinforcing feedback):

- **Worker housing (Zone C)** → requires water, power (lighting, cooking), and healthcare.
- **Workforce availability** → depends on housing conditions, healthcare access, and *predictability of work schedules* (which depends on power).
- **Attrition** → reduces skilled operators → increases equipment mishandling → increases downtime → reduces production → reduces revenue → reduces maintenance budget → increases power failures → further attrition.

Failure propagation path (explicit): Power failure (event) → Equipment unprotected shut-down → Mechanical stress → Cumulative wear → Downtime rises → Production schedule becomes unpredictable → Workers cannot plan shifts → Frustration + income variability → Attrition → Remaining workers overworked → More operator errors → More equipment damage → Power demand fluctuations (due to erratic restarts) → Further power instability.

Missing information in this map: We have no edge weights. How many minutes of power outage cause 1% attrition? How many equipment restarts cause 1% downtime increase? Without these, the map is topological but not operational.

4.2 Binding Constraint

Binding constraint: *Reliability of power delivery as perceived by operators and workers* (not just uptime percentage).

Why it is binding (not just largest problem):

- Resolving attrition (67%) without fixing power would fail because workers would return to an unstable environment.
- Resolving logistics (110 vs 600 trucks/day) without power fails because manufacturing cannot produce output to ship.
- Resolving medical centre overload without power fails because power outages disable medical equipment and refrigeration.
- Resolving water supply without power fails because pumps require power.

In contrast, resolving power reliability to 95%+ uptime with *predictable* outage patterns (scheduled, notified) would:

- Reduce equipment downtime directly (fewer unplanned stops).
- Enable predictable shift scheduling → reduce attrition (workers can plan).
- Restore manufacturing confidence → attract the 4 never-launched units.
- Reduce medical centre surge (fewer heat strokes, fewer accidents from fatigue).
- Stabilise data centre → enable telemetry for further optimisation.

The constraint is binding because it sits upstream of every other flow. Until power is stabilised, all other interventions are palliative.

4.3 Workforce Attrition – Cause, Effect, or Both?

Both, with a reinforcing feedback loop.

Attrition as effect (of power instability):

Unreliable power $\rightarrow$ unpredictable production schedules $\rightarrow$ workers cannot plan overtime or personal life $\rightarrow$ loss of predictable income $\rightarrow$ voluntary attrition. This is the primary direction: 67% attrition is too high to be explained by wages alone given industry norm of 18%.

Attrition as cause (of further failures):

High attrition $\rightarrow$ loss of experienced equipment operators $\rightarrow$ remaining operators run machines suboptimally (e.g., improper shutdown procedures) $\rightarrow$ increased wear $\rightarrow$ higher equipment downtime $\rightarrow$ more production loss $\rightarrow$ more revenue loss $\rightarrow$ less budget for maintenance $\rightarrow$ more power failures (e.g., neglected substation components) $\rightarrow$ even more attrition.

The feedback loop (explicit):

Power uptime ↓ → Production predictability ↓ → Worker frustration ↑ → Attrition ↑
 ↑ ↓
 Maintenance quality ↓ ← Revenue ↓ ← Production ↓ ← Equipment downtime ↑ ← Operator skill ↓

Operational implication: An intervention that only reduces attrition (e.g., wage increase) without improving power reliability will see the effect decay because the causal driver remains. Conversely, improving power reliability alone will reduce attrition but with a lag (workers need to observe sustained stability). The loop must be broken at the power node.

5 DIAGNOSIS

5.1 Three Hypotheses for Medical Centre Overload (340% during shift changes)

Hypothesis	Mechanism	Data to collect
H1: Occupational acute illness surge	Shift changes coincide with peak ambient temperature (semi-arid coastal region). Workers leaving hot manufacturing floors experience heat exhaustion or syncope. Combined with power outages disabling ventilation, heat stress accumulates.	Time-stamped temperature logs in manufacturing zones + medical centre triage records with primary diagnosis codes. Confirm correlation between heat-related diagnoses and shift change times.

H2: Lack of primary care – deferred demand accumulation	The medical centre is the only facility within 38 km. Workers postpone non-urgent care until shift changes (when they have time off). The 340% figure represents <i>outpatient visits</i> (not inpatient beds) during a 2-hour window. Design capacity (120 beds) is irrelevant if measuring footfall.	Collect patient flow data: distinguish bed occupancy from walk-in consultations. If bed occupancy is <100% but footfall is 340%, hypothesis confirmed.
H3: Census undercount – undocumented residents	Actual population served is larger than 4,200. Workers may have families not counted in housing data, or nearby informal settlements use the facility. The 340% surge occurs when these external populations seek care after work shifts elsewhere.	Conduct a 7-day population census at the medical centre entrance (age, residence location, employment status). Compare to official resident registry.

5.2 Distinguishing “Attempted then shut down” vs “Never launched”

What distinguishes them (systemic perspective):

Failure mode	Likely barrier	Implication for NIT’s failure
Never launched (4 units)	<i>Anticipated</i> unreliability. Investors or operators performed due diligence, identified power/water/data risks, and decided not to commit capital. The barrier is <i>ex ante</i> confidence.	Suggests that the <i>reputation</i> of NIT’s infrastructure is already damaged. Even if physical fixes are made, relaunching these units requires credibility restoration, not just technical uptime.
Attempted then shut down (3 units)	<i>Experienced</i> unreliability. They launched, incurred operating losses (e.g., spoiled batches, missed delivery penalties, worker strikes), and exited. The barrier is <i>ex post</i> viability.	Suggests that the failure is not just perception but actual operational collapse. These units provide the most valuable data: their shutdown sequence (power → quality → order cancellation) would reveal the critical threshold of unreliability.

What this distinction suggests about NIT’s failure more broadly:

The failure is *sequential* and *accumulating*. First, the 4 never-launched units indicate that even

before operations began, the expected reliability was below a threshold. Then, the 3 attempted-and-shut-down units demonstrate that the actual reliability was even worse than expected. This implies a *downward credibility spiral*: each failure raises the perceived risk for remaining potential entrants. The township may be in a trap where fixing infrastructure is necessary but not sufficient—there must also be a demonstrable period of stability before new units commit.

5.3 Second- and Third-Order Consequences of 71% Power Uptime

Chain of propagation (explicit, non-summarised):

First-order (direct):

- Power uptime 71% → Manufacturing units experience 29% unplanned downtime.

Second-order (downstream from first-order):

- Manufacturing downtime → Equipment remains in partially completed states (e.g., moulds half-filled, furnaces cooled) → Rework or scrap → Material yield drops → Cost per unit rises.
- Manufacturing downtime → Production schedules become stochastic → Just-in-time supply to logistics hub fails → Trucks arrive but nothing to load → Logistics throughput drops from 600 to 110 trucks/day.
- Manufacturing downtime → Workers are sent home mid-shift (no pay) or held in uncertainty → Predictable income disappears → Workers seek alternative employment → Attrition rises from 18% to 67%.

Third-order (downstream from second-order):

- Logistics throughput drop → Warehouses accumulate empty space → Logistics operators renegotiate contracts or exit (one already intends to exit) → Remaining logistics costs spread over fewer units → Manufacturing logistics cost per unit rises → Manufacturing margins compress further.
- Attrition rise → Remaining workers operate equipment without adequate rest → Error rate increases → Equipment damage accelerates → Equipment downtime rises from 34% (observed) toward higher levels → More production loss even when power is available (feedback loop).
- Attrition rise → Knowledge loss about equipment maintenance schedules (e.g., lubrication intervals, calibration) → Maintenance quality declines → Power distribution equipment (substation, transformers) is not serviced → Power failure frequency increases → Power uptime falls below 71% (observed is average over 12 months; trend may be worsening).

Fourth-order (systemic collapse indicator):

- Margin compression + rising attrition + logistics unreliability → The 3 units that shut down made this calculation explicit. The 4 never-launched units made it implicitly.

Metric cross-propagation table (from power to others):

Power uptime (71%)	Direct effect	Second-order	Third-order	Observed metric
↓	Equipment unprotected stops	Rework, scrap	Margin ↓	Revenue 11cr vs 18cr cost
↓	Unpredictable shifts	Workers leave	Skill loss	Attrition 67%
↓	No output for trucks	Logistics utilisation ↓	Operator exits	Logistics 110/600 trucks/day
↓	Water pumps stop	Water supply failure	Health (dehydration)	Water failure events (0-223)
↓	Data centre instability	No visibility	Cannot diagnose	3G connectivity in 2/4 sub-zones

6 INTERVENTION DESIGN

6.1 Prioritisation Framework

Framework: *Dependency-Weighted Recovery Impact (DWRI)*

Variables (scored 1–5 for each candidate intervention):

1. **Upstream criticality (U):** How many other metrics depend on this component? (Graph out-degree)
2. **Recovery leverage (L):** Expected % improvement in system-wide output per unit spend.
3. **Implementation feasibility (F):** Within 18 months given 45cr. (Inverse of time/cost)
4. **Risk reduction (R):** Reduction in variance of outcomes (stability, not just mean).

Weighted score = $0.4 \times U + 0.3 \times L + 0.2 \times F + 0.1 \times R$

(Upstream criticality weighted highest because binding constraint logic dominates.)

Scoring (illustrative for top candidates):

Intervention candidate	U (1-5)	L (1-5)	F (1-5)	R (1-5)	DWRI	Rank
Power reliability (physical + monitoring)	5	5	3	4	4.5	1
Predictive maintenance platform	4	4	4	4	4.0	2
Workforce attrition analytics & retention	3	3	5	3	3.4	3
Operational command centre (dashboards)	3	3	5	4	3.5	2 (tie)
Medical demand forecasting	2	2	5	2	2.5	4

New manufacturing re-cruitment	2	2	3	2	2.2	5
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Resulting priority order:

1. Power reliability (physical fixes + telemetry)
2. Predictive maintenance platform (enabled by power telemetry)
3. Operational command centre (tied, but sequenced after telemetry exists)
4. Workforce analytics
5. Medical demand (deprioritised – see 6.4)

6.2 What You Would Actually Do (Specific Interventions)

6.2.1 Intervention 1: Power Stability & Real-Time Telemetry (Months 1–6)

Element	Specification
Objective	Increase power uptime from 71% to 92% within 12 months, with <i>predictable</i> outage patterns (scheduled, notified 24h in advance).
Operational problem addressed	Unplanned outages cause cascading equipment damage, attrition, and logistics failure.
Root cause targeted	Lack of condition monitoring on substation components + no early warning system.
Scope	Zone D substation (transformers, switchgear, breakers) + feeders to Zones A, B, C.
Implementation approach	(a) Deploy IoT sensors (voltage, current, harmonic distortion, temperature) on 12 critical nodes. (b) Install cellular backup for data transmission during power loss. (c) Establish maintenance contract with OEM for substation overhaul (8cr). (d) Implement real-time alerting when parameters deviate from baseline.
Budget allocation	14 crore: Sensors & gateways (2cr), Substation overhaul (8cr), Cellular backup (1cr), Installation & commissioning (2cr), Contingency (1cr).
Timeline	Month 1-2: procurement & contracts. Month 3-4: installation. Month 5-6: calibration & baseline collection.
Dependencies	Telecom dispute resolution (need reliable data link) – allocate 1cr to interim 4G routers as bypass.
Expected measurable impact	Uptime → 85% by month 8, 92% by month 14. Outage frequency reduction from 10/week to 2/week. Average outage duration from 45 min to <15 min.
Risks	Sensor supply chain delays; contractor quality. Mitigation: dual-source sensors; third-party inspection.
Monitoring metrics	Uptime % (rolling 7d), outage count/week, mean time between failures (MTBF), mean time to repair (MTTR).

6-month validation	At month 6, uptime 78%? If not, escalate substation replacement instead of repair.
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6.2.2 Intervention 2: Predictive Maintenance Platform (Months 3–12)

Element	Specification
Objective	Reduce equipment downtime from 34% to 15% by predicting failures 48-72 hours in advance.
Operational problem addressed	Unplanned equipment stops (compounding power failures) destroy production schedules.
Root cause targeted	No condition-based maintenance; only reactive repairs after failure.
Scope	All 5 active manufacturing units + critical equipment in Zone D (transformers, pumps, chillers).
Implementation approach	(a) Attach vibration sensors (accelerometers) to 25 critical rotating machines (motors, compressors). (b) Collect 3 months of baseline data. (c) Train XGBoost and Random Survival Forest models with features: vibration spectral bands, power quality (from Intervention 1), operating hours, operator shift. (d) Deploy model with daily risk scores (low/medium/high) for each asset. (e) Integrate with CMMS (computerised maintenance management system) for work order generation.
Budget allocation	11 crore: Sensors & edge gateways (3cr), Data engineering pipeline (3cr), Model development & validation (2cr), CMMS integration (1cr), Training (1cr), Contingency (1cr).
Timeline	Month 3-4: sensor installation. Month 5-7: baseline collection. Month 8-10: model training & validation. Month 11-12: pilot deployment.
Dependencies	Requires stable power telemetry (Intervention 1) for feature engineering. Also requires at least 3 months of equipment run data.
Expected measurable impact	Downtime → 25% by month 12, 15% by month 16. False alarm rate <20% (i.e., 80% of high-risk alerts lead to actual failure within 72h).
Risks	Insufficient failure history for training (only 12 months of data). Mitigation: use semi-supervised anomaly detection (Isolation Forest) as fallback.
Monitoring metrics	Model precision@k (top 10 alerts), recall of failures, mean time to detect (MTTD), maintenance cost per unit.

6-month validation	At month 6 (before model deployment), sensor data collection is on track? Vibration baselines established for 20 assets? If not, extend collection phase.
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6.2.3 Intervention 3: Operational Command Centre & Decision Support (Months 6–14)

Element	Specification
Objective	Provide real-time visibility across all zones to enable coordinated response, not siloed fixes.
Operational problem addressed	Data exists but is not aggregated; no one can see that power failure → water failure → medical surge simultaneously.
Root cause targeted	Lack of centralised operational data backbone (designed but not utilised).
Scope	All zones (A, B, C, D). Single dashboard accessible to governing authority and zone operators.
Implementation approach	(a) Build data lake (cloud-based) ingesting: power telemetry, equipment sensors, water flow meters, medical centre admission logs, workforce attendance, logistics gate transactions. (b) Create dependency graph visualisation (Sankey diagram of failure propagation). (c) Implement anomaly detection (rolling Z-score) on 20 key metrics with SMS/email alerts when 3-sigma deviations occur. (d) Establish daily operations review meeting using dashboard as single source of truth.
Budget allocation	7 crore: Cloud infrastructure (2cr/year), Dashboard development (Power BI embedded, 2cr), Anomaly detection pipeline (1cr), Training & change management (1cr), Contingency (1cr).
Timeline	Month 6-8: data pipeline & lake. Month 9-11: dashboard & alerts. Month 12-14: training & rollout.
Dependencies	Requires telemetry from Interventions 1 & 2. Also requires telecom reliability (allocate 1cr to dedicated fibre line for command centre).
Expected measurable impact	Reduction in mean time to detect (MTTD) outages from hours to <15 minutes. Increase in % of incidents where root cause identified within 4 hours from <10% to >70%.
Risks	Operator resistance to monitoring. Mitigation: design dashboards for operators (show real-time production status, not just managerial KPIs).
Monitoring metrics	Dashboard daily active users, incident detection lag, % of alerts acted upon within 1 hour.

6-month validation	At month 6, data lake ingests 15 data streams? Alerts generated for known past failures? If not, revisit data integration contracts.
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6.2.4 Intervention 4: Workforce Attrition Early Warning System (Months 9–15)

Element	Specification
Objective	Reduce attrition from 67% to 35% within 18 months by identifying at-risk workers before they leave.
Operational problem addressed	Attrition is treated as HR metric, not operational signal. No leading indicators.
Root cause targeted	Lack of real-time worker sentiment and absenteeism tracking.
Scope	All 4,200 residents in Zone C + shift workers.
Implementation approach	(a) Anonymised daily attendance and overtime logs. (b) Weekly pulse survey (3 questions: “Was your shift predictable this week?” “Did you have water/power at home?” “Would you recommend NIT to a friend?”) via SMS. (c) Build logistic regression (interpretable) and random forest (performance) models to predict voluntary exit within next 30 days. Features: absenteeism rate in last 14 days, overtime variance, survey responses, power outage frequency at home zone. (d) Generate daily risk scores: flag workers in top 10% risk for manager check-in.
Budget allocation	4 crore: Survey platform & SMS gateway (0.5cr), Data integration (1cr), Model development (1cr), Manager training for check-ins (1cr), Contingency (0.5cr).
Timeline	Month 9-10: survey deployment. Month 11-12: baseline attrition data. Month 13-15: model & integration.
Dependencies	Power stability (Intervention 1) reduces false positives (workers leaving for reasons outside NIT’s control). Also requires workforce attendance data digitisation.
Expected measurable impact	Attrition → 50% by month 15, 35% by month 18. Precision of exit prediction: 40% (i.e., 40% of flagged workers actually leave within 30 days – high for human behaviour).
Risks	Low survey response rate. Mitigation: incentive (meal voucher for completion). Privacy concerns: anonymise at source.
Monitoring metrics	Survey response rate, model AUC-ROC, % of flagged workers who stay after manager intervention.
6-month validation	At month 9 (before survey launch), baseline absenteeism data available? If not, prioritise digitisation of gate entry logs.

6.3 Sequencing

Order and dependency logic:

Month 1-6: Intervention 1 (Power telemetry + physical fixes)

↓ (provides data foundation)

Month 3-12: Intervention 2 (Predictive maintenance) - overlaps, but full deployment waits for

↓ (provides visibility)

Month 6-14: Intervention 3 (Command centre)

↓ (provides retention tool)

Month 9-15: Intervention 4 (Attrition warning)

Why this order matters:

- **Intervention 1 must come first** because without power reliability, predictive models will be overwhelmed by power-induced failures (signal-to-noise too low). Physical fixes also take longest lead time.
- **Intervention 2 starts early** (sensor installation at month 3) but model training only becomes reliable after power stabilisation (month 8+). Parallel work avoids delaying the platform.
- **Intervention 3 cannot precede data streams** from Interventions 1 & 2. A dashboard with no data is theatre.
- **Intervention 4 is last** because attrition is a lagging indicator. Attempting to predict attrition before power is stable would generate predictions that are correct (workers will leave) but not actionable (nothing to fix except power). Sequencing after power stability ensures that alerts correspond to residual causes (e.g., supervisor quality, wages) that *can* be addressed.

Quick win embedded: Within Intervention 1, deploy cellular-backed alerts for substation parameters in month 4 – immediate reduction in outage duration (knowing which breaker tripped reduces MTTR from 2 hours to 20 minutes).

6.4 What You Are Not Doing

Deprioritised item	Why not addressing now	Opportunity cost
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Medical centre capacity expansion	The 340% figure is likely a <i>scheduling</i> or <i>measurement</i> artefact (see 5.1). Expanding beds without understanding demand pattern risks spending 5-10cr on underutilised assets. Also, medical overload will naturally reduce as power stability reduces occupational injuries (heat stress, accidents from fatigue).	If the hypothesis is wrong and true capacity deficit exists, workers will continue suffering. Mitigation: allocate 0.5cr to a 2-week demand study (Section 5.1 data collection) before committing to expansion. If study confirms H1 or H2, no expansion needed.
Water supply infrastructure upgrades	Water failures (0-223 events) are ambiguous: the range suggests either measurement error or seasonal pattern. Moreover, water treatment depends on power – fixing power may resolve water failures (pumps stay operational). Investing in new wells or pipelines before power would be throwing good money after bad.	If water failures continue after power stabilisation (month 8+), then address in a second 12-month phase. The 45cr does not permit both simultaneously.

Explicit trade-off statement: I am choosing to accept continued medical centre overcrowding and water unreliability for the first 12 months in exchange for fixing the binding constraint (power). If power stabilisation fails to reduce medical load or water failures by month 12, I will reallocate unspent contingency (3cr remaining) to those domains in months 13-18.

6.5 Discipline-Specific Depth (Data Science Engineer)

Problem component tackled: Real-time anomaly detection for power and equipment failures *before* they cause downtime, using only telemetry data without requiring labelled failure events.

Why this is a data science problem, not just an instrumentation problem:

In NIT's current state, we have no historical labels of "failure" vs "near-miss". We cannot train a supervised classifier. However, we can exploit the fact that power quality parameters (voltage, current, harmonics) have a *normal operating region* defined by design specifications. Deviations are anomalies. The challenge is distinguishing *benign transients* (e.g., motor startup inrush current) from *pre-failure patterns* (e.g., harmonic distortion indicating capacitor bank degradation).

Technical architecture:

Sensor layer (vibration, voltage, current, temp)

↓ (edge processing: 1kHz sampling, 10-second aggregates)

Feature extraction (sliding window of 60 seconds)

- Statistical: mean, variance, skew, kurtosis, zero-crossing rate
- Spectral: FFT magnitude at 50/60Hz harmonics (2nd, 3rd, 5th)
- Temporal: autocorrelation at lag 1-10, rolling z-score

↓

Model 1 (unsupervised): Isolation Forest per asset

- Trained on first 7 days of clean data (after substation overhaul)
- Outputs anomaly score (0-1) every minute
- Threshold set at 99th percentile of training scores → triggers alert
- Advantage: no labels needed; detects novel failure modes
- Disadvantage: cannot distinguish failure type

↓

Model 2 (semi-supervised): One-Class SVM on spectral features

- Separately for each asset type (motor, transformer, pump)
- Captures domain-specific normal behaviour (e.g., motors have higher variance during acceleration)
- Outputs binary (normal/anomalous)

↓

Ensemble decision rule:

Alert if (Isolation Forest score > 0.99) AND (One-Class SVM = anomalous) AND (duration > 10s)

- Reduces false positives from brief transients

↓

Post-processing: assign severity level

- Level 1 (alert): anomaly detected, log for review
- Level 2 (warning): 3 anomalies in 10 minutes → SMS to maintenance
- Level 3 (critical): anomaly + power quality degradation (harmonics >5%) → automated equipment shutdown

Validation strategy without labelled failures:

Approach	Implementation
Historical replay	Use first 30 days of data as “presumed normal” (after physical fixes). Then test on next 30 days. Flag any period where equipment actually failed (from maintenance logs) – count as true positive if model alerted within 60 minutes prior.
Simulated injection	Introduce synthetic anomalies (e.g., add 10% harmonic distortion to voltage signal for 5 seconds) during low-production hours. Verify detection within 1 minute.
Operator feedback loop	For each alert, maintenance logs whether inspection found a genuine issue. Compute precision weekly. If precision falls below 30%, raise anomaly threshold.

Alerting architecture (operational integration):

Detection (edge gateway, latency <10s)

↓

Cloud aggregation (5-minute batches)

↓

Severity classification (rule-based + model ensemble)

↓

Notification routing:

- Level 1: Log only (dashboard visualisation)
- Level 2: SMS to shift supervisor + WhatsApp to maintenance lead
- Level 3: SMS to zone head + automated work order in CMMS

↓

Escalation if no acknowledgement within 15 minutes (Level 2) or 5 minutes (Level 3)

Expected operational improvement specific to NIT:

Currently, equipment downtime is 34%, but we do not know what fraction is *preventable* with 48-hour warning. Industry benchmarks for manufacturing with predictive maintenance suggest 40-60% of downtime can be eliminated (reduction from 34% to 15-20%). For NIT, the binding constraint is power, but predictive maintenance on equipment (which suffers cumulative damage from power events) is the *second-order* lever. The data science contribution is to quantify that leverage: after power stabilisation, how much additional uptime can we gain from predicting equipment failures?

Measurable output for this component:

By month 14, a deployed anomaly detection system that achieves:

- Precision (Level 2+ alerts leading to actual maintenance finding) 50%
- Recall (failures preceded by alert within 60 min) 70%
- False positive rate (alerts with no finding) 2 per day per asset

Why a Data Science Engineer approaches this way:

A mechanical engineer would focus on vibration thresholds (e.g., ISO 10816-3) and physical root cause. That is valid but requires labelled failure modes for each asset. A data scientist starts from the *absence of labels* and builds unsupervised methods, then layers interpretability (e.g., SHAP on the One-Class SVM to identify which harmonic caused the anomaly). The trade-off is higher false positives initially, compensated by rapid retraining as maintenance logs accumulate. This approach recognises that in a failing system like NIT, you cannot wait six months to collect labelled failure data – you must detect anomalies from day one.

7 TESTING YOUR OWN PLAN

7.1 Single Most Dangerous Assumption

Assumption: The power substation's components are *repairable* within the 8cr allocated, and the root cause is wear/neglect, not fundamental undersizing.

Why this would undermine the plan: If the substation was undersized from the start

(designed for 125 units but only 5 operational? No – power demand is lower now, not higher), or if the fault is in the grid supply (utility-side), then throwing 8cr at repairs will not improve uptime. The entire cascade of interventions (predictive maintenance, command centre, attrition reduction) depends on power stabilisation. Without it, we spend 45cr and end at 71% uptime.

Early signal (month 4): After substation overhaul and sensor deployment, if uptime does not improve to 78% by month 4 (allowing 2 months for repairs to stabilise), the assumption is wrong.

Contingency (month 4):

- Immediately commission a third-party power systems audit (0.5cr from contingency) to distinguish undersizing vs utility issue vs internal distribution fault.
- If undersizing: replace two largest transformers (6cr additional) – reallocate from Intervention 4 (attrition system delayed by 2 months).
- If utility issue: engage regulator and negotiate dedicated feeder (3cr for political/legal costs) – delay Intervention 2 (predictive maintenance) by 1 month.
- If internal distribution (wiring, breakers): reallocate 2cr from medical study (deprioritise further).

7.2 “New Single Point of Failure” Critique

How this critique applies to my plan:

My plan centralises operational decision-making into the Command Centre (Intervention 3) and relies on real-time telemetry from Interventions 1 & 2. If the command centre’s data pipeline fails (cloud outage, fibre cut, cyberattack), or if the telecom dispute remains unresolved, then all monitoring collapses. Operators revert to blind operation, and the predictive maintenance system becomes silent. This creates a *new* single point of failure: the observability system itself.

Is the critique valid?

Partially yes. The critique is valid if we treat the command centre as a *control* system rather than a *decision support* system. My plan does not automate equipment shutdown (except Level 3 alerts, which are locally stored in edge gateways). Operators can still run equipment without the dashboard, but they lose early warning. The true single point of failure is the *telecom link* from Zone D to the cloud. I mitigated this with cellular backup, but if both fibre and cellular fail (e.g., power outage affecting cell towers), visibility is lost.

Honest response: The critique is valid for months 6-10 during command centre rollout. After month 10, I would implement a local backup: edge gateways store 7 days of telemetry locally and serve a basic dashboard (read-only) even without cloud connectivity. This makes the system resilient to cloud/network failure, shifting the single point of failure to the edge gateway hardware (which has redundancy – dual gateways per zone). No plan eliminates all single points within 45cr; the trade-off is accepting telecom risk in exchange for centralised analytics.

7.3 Contractor Failure at Month 6

Scenario: Two interventions on track (power telemetry, predictive maintenance sensors). One stalled (substation overhaul contractor failed – e.g., went bankrupt, supplied faulty transformers).

Information needed before decision:

1. **Percentage complete on stalled workstream:** Are transformers already installed? Is the contractor's failure permanent or temporary (e.g., cash flow)?
2. **Criticality of stalled workstream to other interventions:** Can predictive maintenance proceed without substation overhaul? (No – if transformers are faulty, sensors will detect anomalies but cannot fix them.)
3. **Sunk cost:** How much of the 8cr already spent? Is recovery from contractor possible (legal, insurance)?
4. **Alternative contractors:** Lead time to onboard a new substation specialist (typically 2-3 months for mobilisation).

Decision framework (go/no-go):

Condition	Decision	Rationale
Workstream <30% complete, contractor permanently failed	Cut losses	Sunk cost is low; new contractor can start fresh without compatibility issues. Reallocate 2cr from contingency to expedite new contract.
Workstream >70% complete, temporary delay (<4 weeks)	Continue	Wait for contractor to resolve. Use the 4 weeks to advance other workstreams (command centre).
Workstream 30-70% complete, permanent failure	Continue with new contractor but de-scope	Accept that substation overhaul will be partial (e.g., replace only the most critical transformer). Downgrade uptime target from 92% to 85%. Reallocate 3cr from Intervention 4 (attrition system) to cover cost overrun.

Specific to this case: At month 6, the stalled workstream is the substation overhaul. This is *critical* – cannot proceed without it. I would ask the governing authority for a 2-week extension to find a new contractor, but if none available, I would pivot to a distributed solution: install power conditioning units (UPS + harmonic filters) at each manufacturing unit (5cr for 5 units) instead of central substation repair. This is more expensive but bypasses the failed contractor. Trade-off: reduces budget for predictive maintenance (delay by 2 months).

7.4 Budget Reduction from 45cr to 28cr at Month 15 (Three Months Before End)

New constraint: 28cr total, already spent ? Assume by month 15, we have spent 35cr (original burn rate). The cut is effective immediately – we have 28cr remaining for months 15-18? Or total budget reduced to 28cr meaning we must return 17cr? The prompt says “budget has been reduced from 45 crore to 28 crore, effective immediately” at month 15 (with 3 months left). I interpret as: total remaining budget for months 15-18 is (28 - amount spent so far). Assuming we spent 35cr in first 15 months, remaining = 28cr - 35cr = -7cr – impossible. So the cut must apply to *total* budget, meaning we have already overspent. More realistic: we have 28cr total for the 18 months, and at month 15 we have 10cr left (having spent 18cr in first 15 months). I will assume the latter: **10cr remaining for months 15-18.**

What changes:

Original plan (remaining months 15-18)	Revised plan with 10cr
Intervention 4 full deployment (4cr)	Cancelled – attrition system not built. Instead, use manual weekly check-ins by supervisors (low-cost).
Intervention 3 dashboard advanced analytics (2cr)	Scaled back – deploy free/open-source dashboard (Metabase) instead of Power BI. No anomaly detection model (manual threshold alerts only).
Intervention 2 model refinement (1cr)	Delayed – use current model as-is (no retraining).
Medical demand study (0.5cr)	Cancelled – accept medical overload as given.
Contingency (3cr)	Reduced to 0.5cr – accept higher risk.

What does not change:

- Intervention 1 (power telemetry) – already deployed by month 15. No further spending needed.
- Intervention 2 core functionality (sensors + basic anomaly detection) – already running.
- Intervention 3 basic dashboard (read-only visualisation) – already built; only advanced features cancelled.

Rationale: Protect the binding constraint (power) and the predictive maintenance platform that depends on it. Non-essential analytics (attrition prediction, medical forecasting) are sacrificed. The system will still operate at 85-90% uptime (achieved by month 14), but workforce retention will suffer without early warning. Acceptable trade-off for 3 months.

8 OPEN RESPONSE

What NIT tells us about the difference between designing a system and designing a system that works

The engineers who designed NIT were not incompetent. The power substation met benchmarks. The medical centre followed standard formulas. The logistics hub aligned with industry specifications. By every *static* measure, NIT was correctly designed. And yet it is failing catastrophically.

This is because designing a *system* (a collection of components that satisfy individual specifications) is not the same as designing a *system that works* (a collection of components that remain viable under dynamic loading, feedback, and degradation). NIT's designers optimised locally: the substation for power quality, the medical centre for bed-to-population ratio, the data backbone for bandwidth. But they did not design for *propagation*.

Consider the medical centre. At design, 120 beds for 4,200 residents is reasonable (29 beds per 1,000 – generous for a working-age population). The failure is not capacity; it is that the *timing* of demand is locked to shift changes, and shift changes are locked to power outages (workers seek care when production stops). No static formula captures that. The system works only if power is stable; if power fails, the medical centre becomes a choke point not because of its own design but because of a failure two zones away.

The data backbone is even more instructive. It was built to provide visibility, but visibility is useless if no one is looking, or if the people looking cannot act. NIT has data – the fact that the case provides 12 months of operational metrics proves data exists. But the metrics are reported as averages (34% downtime, 71% uptime) that hide the *sequence*. The governing authority knows *that* power is unreliable but not *which breaker trips first* or *whether water failures precede or follow power failures*. The design assumed that collecting data is sufficient; the system fails because the data is not structured for causal inference.

The deepest lesson, however, is about *feedback loops that were designed out*. A resilient system has negative feedback: when power fails, maintenance budgets increase. NIT has positive feedback: when power fails, revenue falls, maintenance budgets shrink, power fails more. The designers assumed stable revenue to fund maintenance – a loop they never closed because they did not anticipate the initial instability. They designed for the equilibrium, not the transient.

NIT's failure is not a failure of engineering specifications. It is a failure of *systems thinking*: the inability to see that a 29% power shortfall does not cause a 29% production loss – it causes a 66% capacity loss, 67% attrition, and a medical centre treating 400 patients during shift changes. Specifications are linear; systems are not. Until designers stop asking “does this component meet its target?” and start asking “what happens to everything else when this component fails?”, we will keep building townships that work on paper and collapse in reality.